H2 — ML + Embedded Track

Member: ______

Your role

You own the deploy pipeline documentation and the golden verification of our exported model. You write the team runbook.

Weekly tasks

Week 1 — Deploy pipeline doc

- Task: Read scripts/fetch_model.sh, scripts/deploy.sh, and scripts/benchmark_device.py. Document the pipeline: model.bin, headers, compile, flash. Verify the SHA-256 pin logic by reading.
- Deliverable: The pipeline doc with a diagram and steps.
- Verify: H4 follows the doc and can run deploy.sh.
- Learn: Neural network basics. Watch the 3Blue1Brown series on neural networks.

Week 2 — Eval harness

- Task: Build the eval harness notebook. It pulls checkpoints from Drive and runs analyze.py per run.
- Deliverable: The eval harness notebook in the shared Drive.
- Verify: The notebook logs the ppl of one completed run.
- Learn: Embeddings and next-token prediction. Explain the tokenizer hash check.

Week 3 — Golden verification

- Task: Verify the exported model with the C runtime (ppl.c, host_verify). Record the max abs diff.
- Deliverable: The golden verification result for our exported model.
- Verify: The diff is 1e-5 or below.
- Learn: The transformer block. Explain why the C runtime and PyTorch must match.

Week 4 — Team runbook

- Task: Write the team runbook: every command, every pitfall, Colab tips.
- Deliverable: RUNBOOK.md in the club repo.
- Verify: A new member can flash a model from the runbook alone.
- Learn: Sampling and generation. Explain why sampling uses temperature.

Week 5 — Serial prompting prototype

- Task: Build the serial prompting prototype firmware: read a prompt from serial, generate text.
- Deliverable: The prototype firmware in the club repo.
- Verify: The board responds to a typed prompt.
- Learn: RoPE and KV cache. Explain why the cache matters at runtime.

Week 6 — Demo interaction

- Task: Build the demo interaction: prompt in, story on the screen.
- Deliverable: The working interaction build.
- Verify: The demo runs 10 minutes without a crash.
- Learn: Fine-tuning. Explain why a fine-tuned model needs no firmware change.

Week 7 — Demo script

- Task: Write the demo script and the user guide for the showcase.
- Deliverable: DEMO_SCRIPT.md in the club repo.
- Verify: E1 can run the demo from the script alone.
- Learn: Prepare your teach-back on sampling and generation.

Week 8 — Showcase ops

- Task: Run the showcase booth: setup, demo, teardown.
- Deliverable: The showcase run report.
- Verify: The demo runs during the whole showcase.
- Learn: Write the "what I learned" page.

Learning path

1. Neural network basics: perceptron, MLP, gradient descent.
2. Embeddings, tokenization, next-token prediction.
3. Transformer block: attention, LayerNorm, residuals, MLP.
4. Loss, perplexity, sampling, generation.
5. Attention math, RoPE, KV cache. PLE paper.
6. Fine-tuning, transfer learning, PTQ.
7. Teach one topic to the team.
8. Write the "what I learned" page.

Resources

- Repo: https://github.com/slvDev/esp32-ai
- scripts/: https://github.com/slvDev/esp32-ai/tree/main/scripts
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi